

Homework assignment 4

Due Date: 10th of December EOD

Course: Stochastic Methods in Mathematical Modelling and AI

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Problem 1 Importance sampling (2 points)

Let X be a standard normal distribution ($\mu = 0, \sigma = 1$). Compute $P(X \geq 2)$ by using importance sampling Monte-Carlo. (Hint: use a sampling distribution with a peak of the density close to $x = 2$). Then try to get this value by generating the values by Box-Mueller transform and rejecting the values for $x \leq 2$. Compare the efficiency of two approaches.

Problem 2 Metropolis-Hastings algorithm (3 points)

By using the Metropolis-Hastings algorithm sample the exponential distribution $P(x) = 2 \exp(-2x)$, $x \in [0, \infty)$. Plot the histogram of sampled variables and superimpose the analytical formula. Plot the autocorrelation of the samples.

Problem 3. Fractional derivatives (1 point)

Using the definition and the properties calculate analytically if possible and plot the corresponding Riemann-Liouville fractional derivatives ${}_0D_x^\alpha$ on the interval $[0, 3]$:

1. ${}_0D_x^\alpha x^5, \alpha = 0.25, 0.5, 0.75$
2. ${}_0D_x^\alpha \exp(2x), \alpha = 0.25, 0.5, 0.75, 1$

Problem 4 (Superdiffusion and fractional Fokker-Planck equations, 3 points)

One of the ways to simulate the superdiffusion is to use the overdamped Langevin equation with the white Lévy noise instead of a usual Gaussian noise, i.e. the random variables are taken from the symmetric α -stable distribution. The family of α -stable distributions is a class of distributions which are produced by a sequence of variables with a diverging second moment (according to the generalised central limit theorem). The symmetric α -stable distribution has a characteristic function $\phi(k) = \exp(-\sigma^\mu |k|^\mu)$, where σ is a measure of a width of the distribution and μ is a power-law exponent. The values from α -stable distributions can be sampled, for instance, according to the equations

$$\xi_\mu = \frac{\sin(\mu\Phi)}{(\cos\Phi)^{1/\mu}} \left(\frac{\cos((1-\mu)\Phi)}{W} \right)^{\frac{1-\mu}{\mu}}, \quad (1)$$

here W is an exponential variable ($P(W > w) = e^{-w}, w \geq 0$) and Φ is uniform on $[-\pi/2, \pi/2]$ (More details you could find in chambers1976.pdf).

The rest of the simulation procedure is identical to the Brownian case (Each jump has the same duration). The corresponding discretisation of the equation is, however, slightly different,

$$x_{n+1} - x_n = (K_\mu \delta t)^{1/\mu} \xi_\mu(n), \quad (2)$$

x_n are the positions, K_μ is a fractional diffusion coefficient with dimension cm^μ/sec and $\xi_\mu(n)$ is the α -stable noise which is delta correlated $\langle \xi_\mu(n) \xi_\mu(n') \rangle = \delta(n - n')$, and samples values according to Eq.

(1). μ is used here instead of α in the problem one to avoid confusion. This equation allows to perform a simulation of the process called as Lévy flights (see also an extra material at the end of the file).

1. Simulate a trajectories for Lévy flights in 1D for the starting position $x_0 = 0$, plot them for $\mu = 0.5; 1; 1.5$ and compare them visually with trajectories for the Brownian motion ($\mu = 2$).
2. Plot the probability distribution functions for $t = 0.1, 1, 10, 100$ for stable exponent $\mu = 0.5; 1; 1.5$ (*Reminder.* $\mu = 2$ corresponds to the Brownian motion and $0 < \mu < 2$ to the scale-free Lévy flights).
3. The same problem can be tackled analytically in a simple case of $U(x) = \text{const}$, i.e. when no external force affects the motion. Check the x asymptotic of the power-law tails of the solution $P(x, t)$,

$$P(x, t) \sim \frac{K_\mu t}{|x|^{1+\mu}}. \quad (3)$$

Problem 5. Volatility Smile in Rough Volatility Model (4 points)

Rough volatility models are a class of stochastic volatility models motivated by empirical findings from high-frequency data: the log-volatility process appears to be much less smooth than in classical models that use Brownian motion and can be well-approximated by a fractional Brownian motion with a small Hurst exponent (often reported around $H \approx 0.1$ at reasonable time scales). This led to the adoption of models where volatility is driven by fractional Brownian noise.

A simple rough volatility model can be defined as follows. Let B_t^H be a fractional Brownian motion with Hurst exponent $H \in (0, \frac{1}{2})$, and define the variance process via a rough log-variance:

$$\begin{aligned} dS_t &= \mu S_t dt + \sqrt{\nu_t} S_t dW_t, \\ \nu_t &= e^{X_t}, \\ X_t &= X_0 + \kappa \int_0^t (\theta - X_s) ds + B_t^H. \end{aligned} \quad (4)$$

where:

- S_t is the asset price at time t ,
- ν_t is the variance at time t and $X_t = \log \nu_t$ is the log-variance,
- μ is the expected return of the asset,
- κ is the mean-reversion rate of the log-variance,
- θ is the long-term average level of the log-variance,
- H is the Hurst exponent of the fractional Brownian motion B_t^H ,
- W_t is a Wiener process (standard Brownian motion).

1. Take the notebook from the seminar and implement numerical simulation of the Rough volatility model Eq. 4. You can use external libraries to simulate fractional Brownian motion. Take $H \approx 0.1$. Tune it's paraments to resemble the Apple stock price from 2020 to 2025. (1 point)

2. Write a function to compute the European call option price with the strike price K and expiration time T using the formula

$$C(K, S_0, T) = \int_{-\infty}^{\infty} \max(S_T - K, 0) \cdot p(S_T, t = T | S_0, t = 0) dS_T,$$

where the integral is computed by averaging over an ensemble of rough volatility model trajectories starting at a point $(S_0, t = 0)$. Compute the option price as a function of strike price K and time to expiration T . You need to plot 2D surface (using, for example, `plot_surface` from `matplotlib`) of the option price on (K, T) -plane. (1 point)

3. The Black–Scholes formula connects the option price $C(S, t)$, the current price of the underlying stock S_t , and the volatility σ of the underlying stock:

$$C(S, t) = S_t N(d_1) - K e^{-r(T-t)} N(d_2),$$

where K is the strike price, r is the risk-free rate, T is the expiration time, and $N(\cdot)$ is the cumulative distribution function of the standard normal distribution. The variables d_1 and d_2 are given by

$$d_1 = \frac{1}{\sigma \sqrt{T-t}} \left[\ln \left(\frac{\hat{S}_t}{K} \right) + \left(r + \frac{\sigma^2}{2} \right) (T-t) \right]$$

and

$$d_2 = d_1 - \sigma \sqrt{T-t}.$$

Given the previously computed surface $C(K, T)$, find the surface $\sigma(K, T)$ that minimizes at every point the difference between the predicted option price by the Black–Scholes formula and the simulated option price. As a result, you will get the implied volatility surface as a function of expiration time and the option strike price. Do you observe a pattern similar to a smile in a K -section $\sigma(K, t = \text{const})$? (2 points)

Theoretical detour. Space-fractional Fokker-Planck equation.

The fractional Fokker-Planck equation for superdiffusion is different from the one for subdiffusive case and reads

$$\frac{\partial P(x, t)}{\partial t} = K_\mu \frac{\partial^\mu P(x, t)}{\partial |x|^\mu}, \quad (5)$$

where $\frac{\partial^\mu}{\partial |x|^\mu}$ denotes the fractional space derivative which is called Riesz-Weyl derivative and can be easily defined through it's Fourier transform

$$\mathcal{F} \left(\frac{\partial^\mu P(x, t)}{\partial |x|^\mu} \right) = \int_{-\infty}^{\infty} e^{ikx} \frac{\partial^\mu P(x, t)}{\partial |x|^\mu} dx = -|k|^\mu P(x, t). \quad (6)$$

Hence in Fourier k -space the solution of Eq. 5 can be readily obtained,

$$P(k, t) = \exp(-K_\mu t |k|^\mu). \quad (7)$$

In the real space the solution looks a bit more involved,

$$P(x, t) = \frac{1}{\mu |x|} H_{1,2}^{2,0} \left[\frac{|x|}{(K_\mu t)^{1/\mu}} \left| \begin{matrix} (1, 1/\mu) & (1, 1/2) \\ (1, 1) & (1, 1/2) \end{matrix} \right. \right]. \quad (8)$$

The latter expression has a power-law Eq. 3 asymptotic dependence on x and t . *Remark.* The consequence of this power-law is a divergence of the mean-squared displacement for Lévy flights.